# PIIxel Leaks: Replication & Extension Study

Project Report

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## Abstract

This report presents our replication and extension of "PIIxel Leaks: Passive Identification of Personally Identifiable Information Leakage through Meta's Pixel". Our goal is to validate the original findings, measure how Meta Pixel adoption has evolved across different Tranco snapshots and explore whether the tracker is also present on scam and suspicious domains. To achieve this, we enhanced the original crawler with multithreading for improved runtime and added deep extraction capabilities to catch hidden pixels that evade traditional detection. We conducted our evaluation on the Top 50,000 domains from the authors' 2024 Tranco list, a Top 50,000 list from 2026 and datasets of scam and malicious domains. Our results confirm that the pervasive privacy risks identified in the original study remain highly stable over time and that adversarial actors actively leverage tracking infrastructure to covertly harvest sensitive user data.

## 1 Introduction

Web tracking technologies are deeply embedded in modern websites and Meta Pixel is one of the most visible examples. Because it can observe page visits, form interactions and other user actions, it is highly relevant to privacy research whenever it is deployed on pages that handle personally identifiable information (PII). The original study demonstrated that this is not a theoretical concern: tracking infrastructure can become a channel for unintended data disclosure at scale.

This work replicates and extends the original measurement pipeline to comprehensively assess the current state of Meta Pixel deployment and associated PII leakage risks. To evaluate temporal evolution, we analyze the Top 50,000 domains from the authors' original Tranco list alongside a new Top 50,000 Tranco snapshot from 2026. Furthermore, we expand the scope to adversarial contexts by investigating datasets of scam and malicious domains. Crucially, we extended the measurement crawler to uncover hidden pixels deployed via server-side tagging or obfuscation that would otherwise evade observation.

Our findings demonstrate that Meta Pixel remains broadly present across the domains we investigate and that a substantial fraction of detected deployments are associated with PII leakage. The temporal comparison reveals that PII leakage rates are remarkably stable. Furthermore, we find that adversarial domains actively utilize tracking infrastructure to collect sensitive user profiles. Specifically, we show that PII leakage through Meta Pixel remains a pervasive issue across all observed web populations. Among mainstream websites, leakage rates are remarkably consistent, with 59.3% of the tracking domains in the 2024 Top 50K snapshot and 59.9% in the 2026 snapshot actively transmitting PII. Furthermore, this behavior extends to adversarial contexts, where 53.1% of tracking-enabled malicious domains and 24.6% of scam domains also leak PII. When leakage occurs, the exposure of sensitive data is particularly concerning: over 99% of leaking mainstream websites transmit email addresses and roughly 95% transmit phone numbers. In other words, the core privacy risk identified by the original work remains observable, highlighting the need for continued monitoring and analysis. These results motivate the detailed investigation that follows, which explores adoption patterns, the mechanisms underlying data leakage, the effectiveness of our extended measurement methodology, and the prevalence of tracking instrumentation across different domains and time periods.

## 2 Background

Meta Pixel is a tracking script used by website owners to measure advertising performance and user engagement. In practice, it can record browsing activity and associate it with identifiable events, which makes it a significant privacy concern when deployed on pages that handle sensitive user data. The original study demonstrated that this instrumentation can lead to the leakage of Personally Identifiable Information (PII), such as email addresses and phone numbers.

The underlying mechanism for this data collection often relies on Meta's Advanced Matching feature. When a website owner enables this tool, the Meta Pixel is instructed to automatically scan the web page for any form fields that look like personal information—such as the email address or phone number entered during a checkout process or registration. Once found, this information is packaged and sent back to Meta's servers. Although websites typically scramble or hash this data before sending it to protect user privacy, security researchers have repeatedly shown that this provides very little protection, as the original information can often be easily guessed or reconstructed. Furthermore, as privacy awareness has grown, many trackers have begun using more sophisticated methods to collect this data. This shift makes it increasingly difficult for ordinary users and traditional measurement tools to see exactly what sensitive information is leaving their browsers.

The web ecosystem is characterized by rapid evolution: domain rankings shift, site ownership changes and deployment practices adapt in response to regulatory pressures and market dynamics. Tracking adoption patterns may vary significantly depending on website category, business model and geographic context. By examining Meta Pixel across multiple temporal snapshots and diverse domain populations, we can establish whether observed privacy risks are endemic to tracking infrastructure or contextual to specific web populations.

# 3 Related Work

The most directly relevant prior work is "PIIxel Leaks: Passive Identification of Personally Identifiable Information Leakage through Meta Pixel", which showed that Meta Pixel can passively observe PII entered into online forms, including email addresses and phone numbers and that such leakage can occur at scale across real websites [1]. Our study replicates and extends this line of work on newer Tranco snapshots and on malicious and scam datasets, allowing us to test whether the leakage patterns reported previously remain present in a newer web population and in more adversarial contexts.

The paper itself builds on earlier work, especially "The Hitchhiker's Guide to Facebook Web Tracking with Invisible Pixels and Click IDs", which examined Facebook's broader tracking ecosystem and showed that invisible pixels and click IDs can be used to follow users across sites [2]. That study established that Facebook tracking is not limited to conventional third-party cookies, but instead relies on multiple instrumentation mechanisms that operate at different points in the browsing and conversion flow.

Crucially, as the advertising ecosystem has evolved in response to browser-level privacy controls, Meta has increasingly incentivized the use of server-side data collection mechanisms. The Conversions API (CAPI) is officially documented by Meta as a direct channel to send web events and user data directly from the advertiser's servers to Meta's infrastructure [5]. While traditional studies of pixel leakage rely exclusively on client-side observation, the widespread adoption of CAPI underscores the critical need for the extended detection capabilities we introduce in this work, as server-to-server data flows are inherently invisible to conventional browser-based measurement.

# 4 Methodology

## 4.1 Parallelized Hybrid Crawler Architecture

Our methodology builds upon the crawler used in the original study, which we modified to support massive parallelism and active evasion techniques. By reorganizing the crawl pipeline with `concurrent.futures`, static HTML fetches run in thread pools while headless browser sessions run in process pools, bypassing the Global Interpreter Lock (GIL) for CPU-bound Selenium instances. We also parallelized the resolution of individual Google Tag Manager (GTM) containers within the same domain to prevent network I/O bottlenecks.

To accurately measure tracking across adversarial environments, our crawler implements automated evasion techniques. For sites protected by Web Application Firewalls (WAF) or bot-mitigation services (e.g., Cloudflare, Akamai), the headless browser strips `navigator.webdriver` signals and applies anti-detection flags to bypass challenge pages. We also implemented a heuristic autoclicker to dismiss GDPR consent banners and "Accept All" cookie walls, ensuring that conditional tracking pixels are successfully triggered. Ultimately, this hybrid design maximized throughput while maintaining the high-fidelity rendering required to detect evasive trackers at scale.

## 4.2 Data Extraction and Validation

After crawling, we extract all pixel-related evidence from the collected artifacts using a multi-layered approach. Many domains, particularly scams, obscure their tracking using Server-Side Tagging (Conversions API) and headless background requests, bypassing conventional front-end extraction. To address this, we extended our crawler with comprehensive deep extraction capabilities. Our network interception engine parses JSON POST bodies to actively capture Meta Conversions API (CAPI) events and expanded cookie detection to include click identifiers (`_fbc`) and encrypted tracking cookies (`fr`). Furthermore, our crawler recursively parses `localStorage` and dynamically injected DOM elements to recover dormant IDs that evade ad-blockers.

Each candidate pixel is then validated by checking its configuration ID against Meta's infrastructure; only IDs that resolve to a valid, active Meta Pixel are retained. Once validated, we record pixel ID reuse across domains to support our clustered infrastructure and PII leakage analysis.

## 4.3 Meta Pixel Adoption

To evaluate the prevalence of Meta tracking across the web, we calculate the Meta Pixel adoption rate within our datasets. We designed this analysis to directly replicate the bucketed adoption measurements from Figure 4 of the original paper. While the original study analyzed the entire Tranco Top 1M list, our replication was limited to the Top 50K domains and segmented them into five rank-based buckets of 10,000 domains each. For each bucket, we compute the percentage of active domains that successfully triggered at least one valid Meta Pixel during our crawl. This strategy preserves the original methodology for comparing tracking density between highly trafficked mainstream sites and the long-tail of the web, forming the basis for the adoption trends presented in our evaluation.

# 5 Datasets

## 5.1 Tranco Top50K snapshots

We evaluate two Top50K Tranco snapshots to measure how Meta Pixel adoption and leakage behavior change over time. The first snapshot is the authors' list from December 2024 and the second is the May 2026 snapshot collected for this study. Running the same pipeline on both datasets lets us compare what stayed stable and what changed in the broader tracking ecosystem across the 18-month gap, using the same crawl and validation procedure.

## 5.2 Malicious and scam datasets

In addition to the general web snapshots, we analyze a malicious-domain dataset from [3]. This dataset lets us examine whether Meta Pixel is reused across suspicious infrastructure and whether the same tracking identifiers cluster across related domains.

We also run the crawl on a set of scam domains drawn from Facebook scam advertisements collected by Norton [4]. This second dataset complements the malicious one by providing a different source of potentially deceptive domains, allowing us to check whether Meta Pixel appears in scam campaigns and whether those sites also expose PII.

# 6 Detection & Adoption Results

## 6.1 Tranco Snapshot Comparison

The two Top50K snapshots capture different web populations separated by roughly 18 months, so comparing them gives us a longitudinal view of how Meta Pixel deployment changes over time. The older list reflects the December 2024 ranking, while the newer list reflects the May 2026 ranking. Although the specific domains in each snapshot differ, the overall level of detection remains remarkably close. Specifically, out of 45,858 successfully processed domains in the 2024 snapshot, we detected Meta Pixel on 12,243 domains (26.7% adoption). In the 2026 snapshot, out of 45,979 successfully processed domains, we detected Meta Pixel on 11,944 domains (26.0% adoption). For comparison, the authors of the original study reported an adoption rate of 24.96% within the Top 10K domains, which aligns closely with our broader observations across the Top 50K. This suggests that Meta Pixel adoption among high-traffic sites is fairly stable even as the ranking set evolves.

**Table 1: Table 1 of the Paper: Distribution of Meta Pixel instance types across the Top50K runs. Percentages are computed over detected instances for the listed categories.**

| | Top50K 2024 (Old) | | Top50K 2026 (New) | |
|---|---|---|---|---|
| **Identification Method** | Websites | % Inst. | Websites | % Inst. |
| Static: GTM | 5,119 | 41.6% | 4,867 | 40.5% |
| Static: Meta Pixel | 1,541 | 12.5% | 1,570 | 13.1% |
| Static: Other Scripts | 337 | 2.7% | 327 | 2.7% |
| Headless: Meta Pixel | 2,522 | 20.5% | 2,508 | 20.9% |
| Headless: GTM | 2,519 | 20.5% | 2,465 | 20.5% |
| Headless: HTML | 110 | 0.9% | 122 | 1.0% |
| Headless: CAPI/Network | 112 | 0.9% | 110 | 0.9% |
| Headless: Deep Ext. | 36 | 0.3% | 36 | 0.3% |

Table 1 summarizes the detection breakdown for the two Top50K snapshots. The overall number of detected domains is very similar across the two runs: 12,243 in December 2024 and 11,944 in May 2026. This corresponds to adoption rates of 26.7% and 26.0%, respectively, suggesting that Meta Pixel remains widely deployed even though the exact set of domains changes over time.

The static and headless proportions are also stable across snapshots. In both runs, roughly 56% of detections come from static crawling (56.9% in 2024, 56.3% in 2026) and 44% from headless crawling (43.1% in 2024, 43.7% in 2026). GTM remains the dominant detection type in both snapshots, but direct Meta Pixel implementations are still substantial: in the 2026 run, 1,570 websites were identified statically with the native Meta Pixel script and 2,508 through headless crawling. Compared with the 2024 snapshot, the 2026 run shows a small decline in total detections, but not a behavioral shift in the type of instrumentation we observe.

Importantly, our extended detection methods contributed additional findings that the original methodology would have missed entirely. Across both snapshots, CAPI/Network interception detected approximately 110 websites whose tracking operated exclusively through server-side Conversions API calls invisible to front-end analysis. Deep Extraction recovered 36 additional websites in the 2024 snapshot and 36 in the 2026 snapshot by parsing `localStorage` entries and inline source-code patterns. While these categories represent a small share of the overall detection volume (roughly 1.2% each), they confirm that a non-trivial number of mainstream Top50K websites have already migrated to obfuscated tracking implementations that evade traditional crawling approaches.

These results indicate that the broader Meta Pixel ecosystem is still large and that the crawl pipeline continues to capture both direct pixel deployments and tag-manager-mediated instrumentation at similar rates across time.

**Table 2: Extension of Table 1 of the Paper: Distribution of Meta Pixel instance types across the malicious and scam runs. Percentages are computed over detected instances for the listed categories.**

| | Malicious | | Scam | |
|---|---|---|---|---|
| **Identification Method** | Websites | % Inst. | Websites | % Inst. |
| Static: GTM | 1 | 1.0% | 1 | 0.2% |
| Static: Meta Pixel | 54 | 55.7% | 136 | 25.5% |
| Static: Other Scripts | 2 | 2.1% | 2 | 0.4% |
| Headless: Meta Pixel | 30 | 30.9% | 379 | 71.0% |
| Headless: GTM | 9 | 9.3% | 5 | 0.9% |
| Headless: CAPI/Network | 1 | 1.0% | 0 | 0.0% |
| Headless: Deep Ext. | 0 | 0.0% | 11 | 2.1% |

## 6.2 Extension: Malicious-domain Analysis

The malicious-domain crawl processed 7,265 domains in total, the vast majority of which returned no pixel evidence (7,168 domains yielded no pixels, including 1,545 with invalid HTML and 2,251 that timed out). Despite this, we detected 97 total pixel instances across 64 validated domains, yielding a detection rate of 0.9% over the full crawl set. Static crawling contributed 57 instances (58.8%) while headless crawling contributed 40 (41.2%).

The pixel-type breakdown reveals that malicious domains rely almost exclusively on direct Meta Pixel implementations rather than tag managers. Of the 57 static detections, 54 were native Meta Pixel (PIXEL_CORE) instances, with only 1 GTM detection and 2 third-party script detections. However, the static validation rate was notably low at 42.1%, suggesting that many malicious sites embed invalid or recycled pixel IDs. In contrast, all 40 headless detections validated successfully (100%), with 30 coming from first-party cookie (FBP) detection, 9 from GTM resolution, and 1 from CAPI/Network interception. The re-use pattern we observed indicates that a relatively small set of pixel identifiers may be deployed across multiple suspicious domains. Taken together, the combination of heavy pixel reuse, near-absent GTM usage, and a high rate of invalid static IDs paints a picture of coordinated, disposable campaign infrastructure that is fundamentally different from the legitimate tracking ecosystem observed in the Tranco snapshots. Furthermore, data collection within this infrastructure is highly aggressive: 34 of the 64 validated malicious domains (53.1%) actively leak PII parameters. The full detection breakdown for both adversarial datasets is presented in Table 2, which directly extends the Top50K analysis from Table 1 and highlights the stark contrast between mainstream and malicious tracking behavior.

### 6.3 Extension: Scam-domain Analysis

To accurately evaluate the adoption rate of tracking infrastructure among scammers, we filtered our initial dataset of 1,000 domains to isolate only the 653 domains that are currently active and serving content.

Within this active subset, the scam-domain crawl shows an extraordinarily strong Meta Pixel presence. We detected 533 total pixel instances, with 474 domains successfully validated, translating to an astonishing adoption rate of 72.6%. Headless crawling dominated the detections here, accounting for 394 instances (73.9%), while static analysis contributed 139 instances (26.1%). Notably, the headless detections achieved a near-perfect validation rate.

The pixel-type breakdown is also striking. Among static detections, 136 were direct Meta Pixel (PIXEL_CORE) instances with only 1 GTM detection. On the headless side, 379 came from first-party cookie (FBP) detection, 5 from GTM resolution, and 2 from deep source extraction. In other words, the scam set is overwhelmingly dominated by direct Meta Pixel rather than GTM or other tracking scripts. This aligns with the broader observation that scam campaigns make active use of tracking infrastructure to aggressively target susceptible demographics, providing a clear setting in which to study the PII leakage behavior discussed later in the report.

Crucially, our Deep Extraction methodology proved its value within this adversarial dataset. We discovered 9 domains actively tracking users through `localStorage`-based storage, effectively hiding their Pixel IDs from traditional observation. All 9 of these deep extraction detections validated successfully, confirming that scam operators are actively deploying tracking obfuscation techniques that only our extended methodology can uncover. Beyond just pixel presence, these scam domains demonstrate significant data collection behavior, with 117 of the 475 validated domains (24.6%) actively leaking PII parameters back to tracking servers.

### 6.4 Clustered Infrastructure Analysis

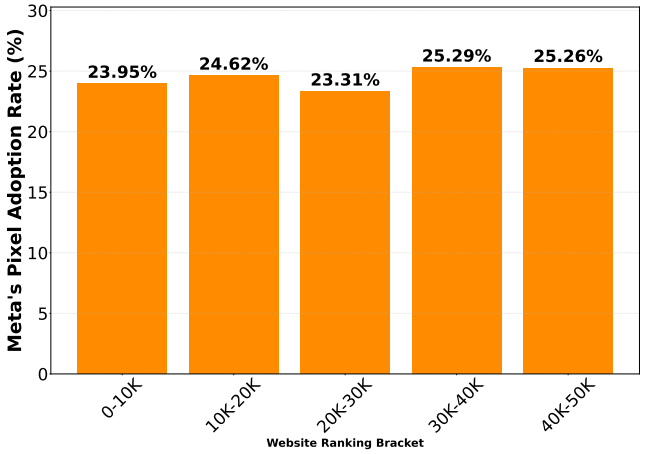
We then examined whether the reused pixel IDs also correspond to shared infrastructure characteristics such as registrar, ASN, and hosting. The malicious set is much more tightly clustered than the scam set: 54 of the 64 validated malicious domains fall into 14 clusters, meaning an extraordinary 84.4% of the list shares a reused pixel with at least one other domain. Most malicious clusters are small pairs, but several are larger groups, including clusters of 3, 5, 6, 8, and 12 domains. Infrastructure overlap is extremely strong at the hosting layer, with all 14 clusters sharing an ASN and 10 clusters sharing a registrar.

In stark contrast, the scam set shows far less clustering. Only 34 of the 474 validated active scam domains (7.2%) share a pixel with another domain in the dataset. These form 18 small clusters—primarily pairs, along with two triplets. However, where clustering does exist in the scam set, infrastructure overlap remains strong, with 14 of the 18 clusters sharing an ASN and 12 sharing a registrar. This suggests that while malicious domains are heavily coordinated in tight networks, scam domains operate much more independently or across wider unobserved networks.

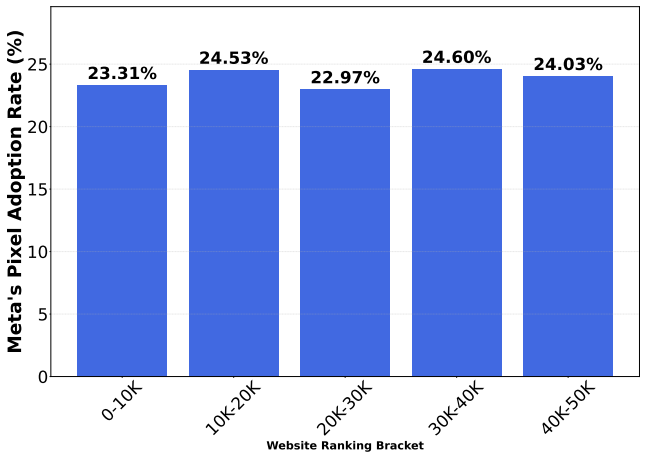
These results mean that reused Meta Pixel identifiers are not appearing on random, unrelated sites. Instead, they often coincide with shared backend infrastructure, especially shared ASN and, in several clusters, shared registrar. This is evidence of operational coordination or campaign reuse, not proof of common ownership, but it strengthens the argument that pixel reuse is linked to highly organized deployment patterns—such as algorithmic domain generation—rather than isolated one-off sites.

## 7 Evaluation

### 7.1 Figure 4: Adoption Across Rankings




**(a) Adoption per rank (Old Tranco)**



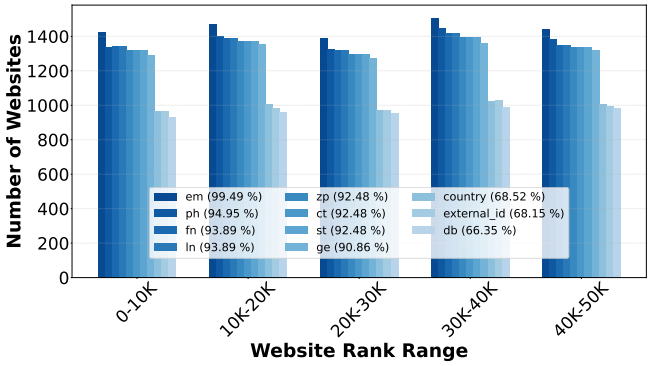

**(b) Adoption per rank (New Tranco)**

**Figure 1: Figure 4 of the Paper: Adoption rates across website ranking buckets (2024 VS 2026).**

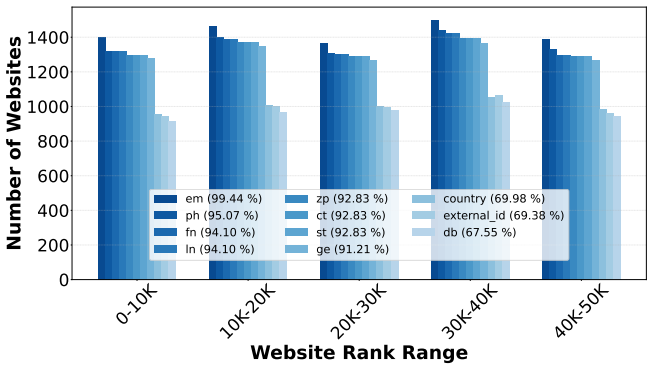
Figure 1 illustrates the adoption rate of Meta Pixel across different domain ranking buckets, comparing the 2024 and 2026 Tranco snapshots. Rather than a steady decline from the top, adoption rates actually oscillate across the buckets, remaining relatively stable between 23% and 25%. In both snapshots, adoption peaks slightly in the 30,000–40,000 rank bucket (reaching roughly 25%), closely followed by the 10,000–20,000 bucket. Conversely, adoption drops to its lowest point (around 23%) in the 20,000–30,000 bucket,

while the absolute top 10,000 domains sit closer to the baseline average. Notably, the overall shape of this oscillating distribution is remarkably consistent between the two time periods, reinforcing the conclusion that Meta Pixel's market penetration is structurally ingrained across the entire Top 50K ecosystem rather than being strictly tied to the very highest traffic volumes.

## 7.2 Figure 5: Parameter Adoption Per Rank




(a) Old Tranco



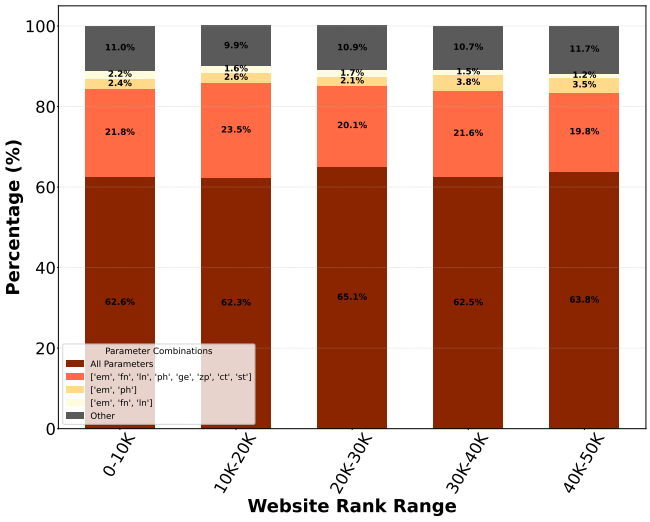

(b) New Tranco

**Figure 2: Figure 5 of the Paper: Parameter adoption per rank batch across different datasets.**

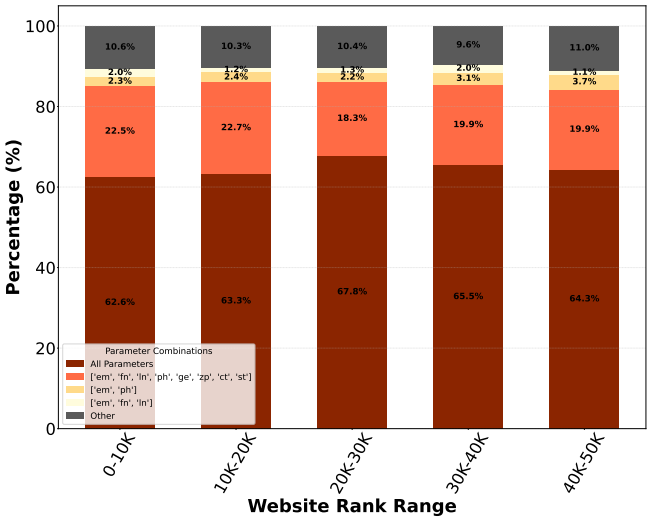
Figure 2 breaks down the adoption of specific Advanced Matching PII parameters across the same ranking batches. Because the overall Meta Pixel adoption rate fluctuates across rank buckets (as seen in Figure 1), the absolute volume of parameter collection naturally mirrors that oscillating trend, peaking in the 30,000–40,000 bucket. However, the *proportional* rate of collection remains surprisingly consistent regardless of rank: whether a site is in the top 10K or the bottom 10K, if it implements tracking, it tends to collect the same comprehensive set of identifiers. Email (`em`) and phone numbers (`ph`) consistently dominate the collection landscape in every bucket.

## 7.3 Figure 6: Parameter Combinations

While the presence of such parameters in a website's configuration is a robust indication of PII tracking, relevancy attribution of ads is a multi-faceted process. Demographics can be used to cluster a group of users based on shared attributes, with the most common being the age, gender and region. Gender and birthday are also required fields when creating a Facebook account, but are not unique to the extent email accounts and phone numbers are. Although regional identifiers are not required, they can be collected through Facebook Location Services if a user enables them. To identify the most commonly tracked groups of parameters, we measure the presence of each parameter combination tracked through the collected configurations.




(a) Old Tranco




(b) New Tranco

**Figure 3: Figure 6 of the Paper: Distribution of parameter combinations across different datasets.**

Figure 3 visualizes the most frequent parameter combinations observed across the different datasets. Across both snapshots, the dominant combinations consistently involve the full identity of the

user, accounting for approximately 64% of all configurations in both the 2024 and 2026 snapshots. The second most frequent is an 8-parameter combination (`ct, em, fn, ge, ln, ph, st, zp`), which makes up another 21% of the dataset. Together, these two extremely comprehensive tracking setups represent 85% of all observed Meta Pixel integrations. By contrast, partial or minimalist tracking—such as collecting only an email address (`em`) or an email and phone number pair (`em, ph`)—accounts for less than 5% of deployments combined. This contrast indicates that websites rarely implement partial or selective tracking; instead, the ecosystem standard is to capture a complete, multi-faceted user profile whenever possible.

## 7.4 PII Leakage

**Table 3: Table 2 from Paper: PII leakage for Top 50k 2024 vs 2026**

| PII | Parameter | 2024 | 2026 | Required |
|---|---|---|---|---|
| Email | em | 99.5% | 99.4% | Yes |
| Phone Number | ph | 94.9% | 95.1% | Yes |
| First Name | fn | 93.9% | 94.1% | Yes |
| Last Name | ln | 93.9% | 94.1% | Yes |
| City | ct | 92.5% | 92.8% | No |
| State | st | 92.5% | 92.8% | No |
| ZIP Code | zp | 92.5% | 92.8% | No |
| Gender | ge | 90.8% | 91.2% | Yes |
| Country | country | 68.5% | 70.0% | No |
| external_id | external_id | 68.1% | 69.4% | No |
| Birthday | db | 66.3% | 67.5% | Yes |

Table 3 details the specific Personally Identifiable Information (PII) parameters exposed through Meta Pixel's Advanced Matching feature, comparing the 2024 and 2026 Tranco snapshots. The most striking finding is the sheer stability of the data collection: the proportion of domains leaking each specific parameter remains virtually identical across the 18-month gap. This longitudinal stability strongly suggests that data-sharing behaviors are deeply structurally embedded within web tracking integrations, rather than being transient configurations.

Uniquely resolvable identifiers remain the primary target of these tracking infrastructures. Email addresses (`em`) are exposed in over 99% of leaking deployments, while phone numbers (`ph`) are exposed in roughly 95% of cases.

Furthermore, the data reveals a systematic approach to profile enrichment. Core identity attributes such as First Name (`fn`) and Last Name (`ln`) are leaked alongside highly localized geographic data (City, State, ZIP) at consistent rates of over 92%. This implies that tracking implementations are not merely capturing isolated fragments of user data; they are designed to vacuum up the entire identity matrix submitted in web forms. Ultimately, these results confirm that the privacy risks identified in the original study have not reduced. Instead, the persistent leakage of highly sensitive, required identifiers remains a fundamental characteristic of the modern web ecosystem.

Beyond mainstream domains, our analysis of adversarial environments reveals that scam and malicious domains actively leverage server-side tagging (e.g., Meta CAPI) and JavaScript obfuscation to silently harvest this same PII while evading traditional detection. The fact that our deep extraction methodology successfully recovered full identity matrices from these obscured implementations highlights that modern privacy threats are deeply embedded and actively evolving to avoid scrutiny.

## 8 Limitations

A primary limitation of our study is compute capability. While the original authors possessed the resources to evaluate the entire Tranco Top 1M list, our local machine constraints limited our crawler to only processing the Top 50,000 domains. As a result, our findings reflect tracking behavior at the top of the web ecosystem, but we cannot comment on the long-tail behavior of lower-ranked sites. Furthermore, if we had the computational resources to rerun the crawler and retry the domains that temporarily hanged or returned invalid HTML during our single execution pass, we likely would have captured an even greater volume of tracking infrastructure and PII leakage.

## 9 Conclusion

In this study, we successfully replicated and significantly extended the baseline methodology of the *PIIxel Leaks* study to evaluate the current state of Meta Pixel tracking. By engineering a highly parallelized hybrid crawler, we drastically reduced execution times, enabling longitudinal analysis across multiple 50,000-domain Tranco snapshots. Crucially, we extended the crawler with advanced capabilities, allowing us to successfully discover and extract hidden pixels that purposefully evade traditional front-end observation.

Our results confirm that the pervasive privacy risks identified in 2024 remain completely unabated in 2026. Meta Pixel adoption continues to closely track with domain popularity and the collection of full identity matrices (emails, phone numbers, names, and locations) remains exceptionally stable across mainstream sites. Furthermore, by applying our extended methodology to adversarial datasets, we demonstrated that scam and malicious domains are actively weaponizing these advanced tracking technologies to covertly harvest sensitive user data. Ultimately, this work proves that as tracking infrastructures become more opaque, measurement methodologies must evolve to hold the ecosystem accountable.

## References

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